Vector Calculus, Linear Algebra and Differential Forms: A Unified Approach - Solutions

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Chapter 0

Preliminaries

0.2 Quantifiers and Negation

Exercise 0.2.1 Negate the following statements:

a. Every prime number such that if you divide it by 4 you have a remainder of 1 is the sum of two squares.

b. For all $x \in \mathbb{R}$ and for all $\epsilon > 0$, there exists $\delta > 0$ such that for all $y \in \mathbb{R}$, if $|y-x| < \delta$, then $|y^2-x^2| < \epsilon$.

c. For all $\epsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in \mathbb{R}$, if $|y-x| < \delta$, then $|y^2-x^2| < \epsilon$.

SOLUTION

a. There exists a prime number in the form of $4n+1$, for $n \in \mathbb{N}_0$ that cannot be expressed as the sum of two squares.

b. There exists $x \in \mathbb{R}$ and $\epsilon > 0$ such that for all $\delta > 0$, there exists $y \in \mathbb{R}$ with $|y-x| < \delta$ but $|y^2-x^2| \geq \epsilon$.

c. There exists $\epsilon > 0$ such that for all $\delta > 0$, there exists $x, y \in \mathbb{R}$ with $|y-x| < \delta$ but $|y^2-x^2| \geq \epsilon$.

0.3 Set Theory